Thermal and Alternating Magnetic Field Demagnetization Experiments on Cainozoic Basalts from Victoria, Australia

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Summary

A re-examination of the palaeomagnetism of the Older Cainozoic Volcanics of Victoria has indicated that these rocks possess temporary unstable components of magnetization. Thermal demagnetization experiments show the magnetic mineral in these basalts is a titanomagnetite. A.c. cleaning experiments to remove the unstable components of magnetization are also described.

1. Introduction

The results of a palaeomagnetic survey of the Older Volcanic basalts of Victoria have previously been reported by Green & Irving (1958) in a paper which also included results from the Newer Volcanic basalts. It was found that while the mean direction of magnetization of the localities sampled in the Newer Volcanics coincided closely with the direction of the Earth's present geocentric axial dipole field the mean inclination of magnetization of localities in the Older Volcanics was 16° greater.

It was thought that because of the nearness of the mean direction of magnetization of these basalts to the present day dipole field, an independent check of the result would useful, for no stability tests were carried out on the rocks, using a.c. demagnetization apparatus.

2. Field work and results

Most of the localities recorded by Green and Irving were resampled together with several new ones. In all, 53 specimens were collected from 21 localities. Mean site directions are shown in Figure 1.

It is seen that there are sites of both normal, reversed and mixed polarity among them as was previously reported by Green and Irving.

Mean site directions are given in Table 4.

The mean direction of magnetization determined from the results of the present investigation are given in Table 1.

Generally speaking these results agree closely with the previous investigation of Green and Irving. Sites exhibiting mixed polarities of magnetization are discussed in Section 8.

* Received in original form 1962 April 9.

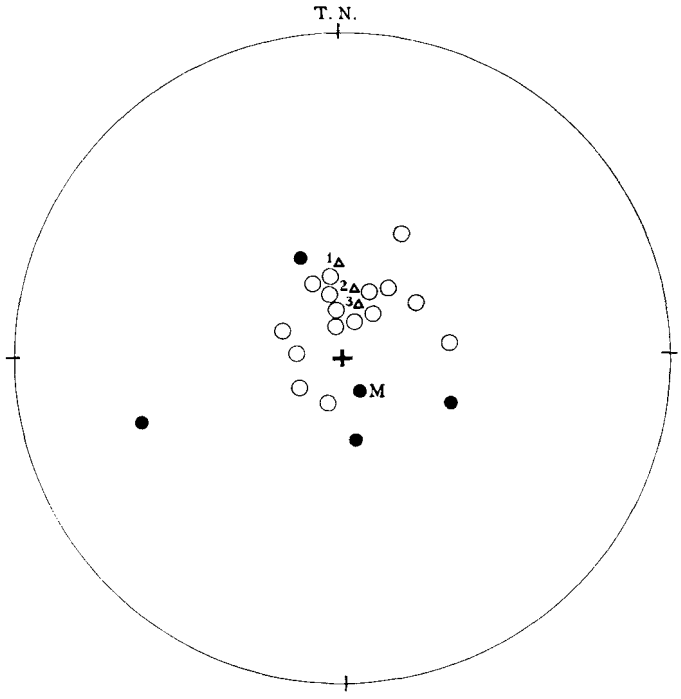


FIG. 1.—Stereographic projection showing mean site directions before a.c. cleaning. Full dots are north-seeking directions plotted on the lower hemisphere (reversed), and the open circles are north-seeking directions in the upper hemisphere (normal). The point *M* is the site of mixed polarity.

△ 1. geocentric axial dipole field
△ 2. present field
△ 3. mean of site directions

Table 1

	N	D(TN) deg.	I deg.	*K*	α deg.	β deg.
Green & Irving (1958)	15	17	−72·9	430·5	6·8	35
Present investigation—before a.c. cleaning	21	18½	−71·5	229	9	11·5
Present investigation—after a.c. cleaning	20	10	−77	257	9	15·5

N is the number of sampling sites. D and I are the declination and inclination. *K* is the overall precision (Watson & Irving (1957)), α is the radius of the Fisher error circle (Fisher (1953)) at $P = 0{\cdot}05$, calculated from a modified formula from Watson and Irving. β is the between site dispersion.

D = 0°, I = −57° for the geocentric axial dipole field.

D = 11°, I = −66° for the present field.

The sampling area for the Older Volcanics of Victoria is 38°·0 s, 145°·5 E.

3. Stability of magnetization

In order to retain some check on the stability of the magnetization in these rocks, the directions of magnetization of many of these basalts were remeasured about nine months after the original measurements. It was observed that in some cases the direction of magnetization of rock disks cut from a particular sample changed quite considerably, that is, by amounts much more than the experimental errors involved.

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In Figure 2(a) are shown the original directions of magnetization of disks cut from specimen 36, together with the directions obtained in re-measurement. Figure 2(b) shows similar results for specimen 19, and Figure 2(c) for specimen 25.

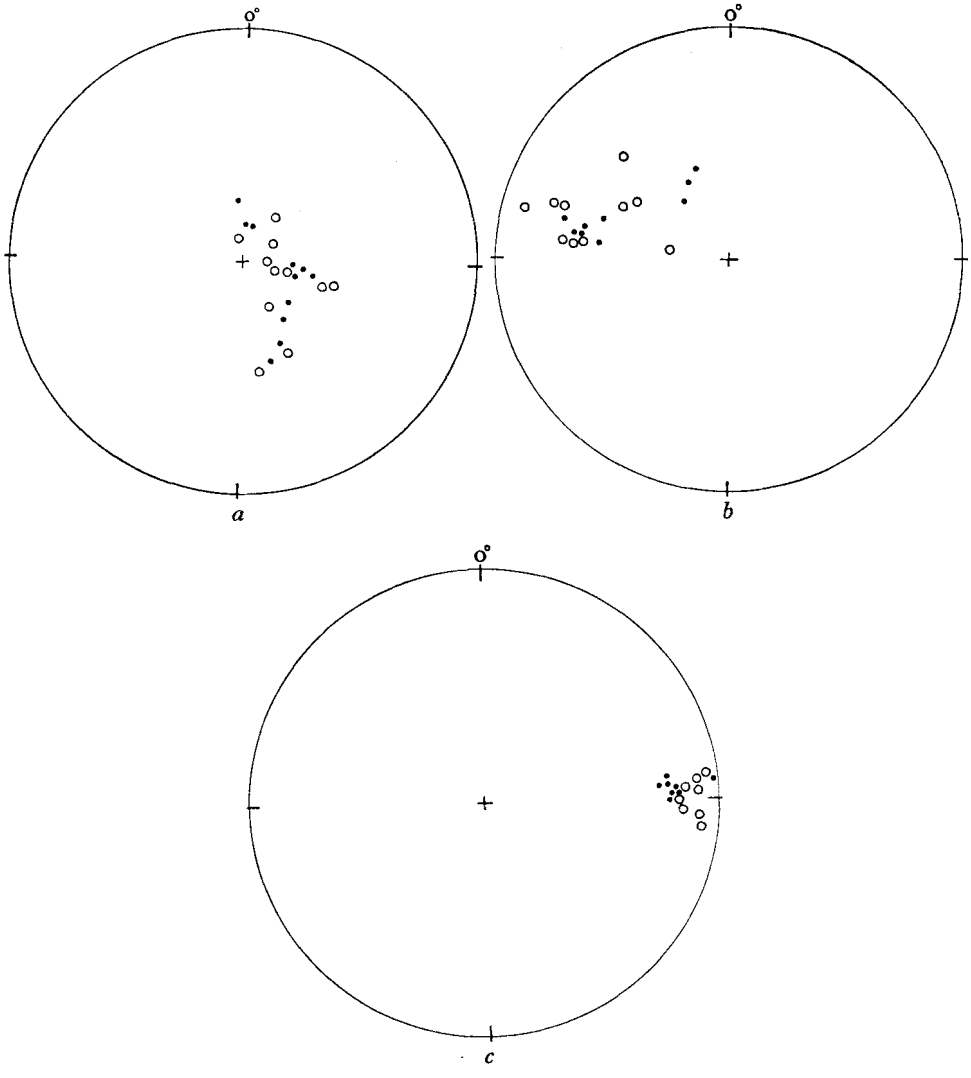


FIG. 2.—Stereographic projection showing directions of magnetization before and after nine months storage at an angle to the Earth's field, for

(a) Specimen 36
(b) Specimen 19
(c) Specimen 25

● north-seeking directions of original measurements.
O north-seeking directions on remeasurement after nine months. Both sets of points are plotted on the upper hemisphere.

The directions of magnetization given in these figures are directions relative to the disks only. They have not been corrected to take into account the orientation of the samples from which they were removed.

It was obvious from such results that there was some measure of instability present and so the thermal demagnetization and a.c. demagnetization experiments described in Sections 4 and 5 were carried out to study this more fully.

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4. Thermal demagnetization experiments

The thermal demagnetization apparatus was built to determine thermal demagnetization curves for basalt specimens. From these curves it was hoped that some definite conclusions could be made about the magnetic minerals in these basalts.

In an earlier paper, Parry (1960) has described experiments recording the change of saturation magnetization with temperature for specimens of basalt taken from the Newer Volcanics. In the following sections thermal demagnetization experiments carried out in zero field on specimens taken from the Older Volcanic basalts are described. These experiments therefore record the change in the natural remanence of the specimens with temperature.

The thermal demagnetization apparatus constructed by the author consists of three main parts.

4.1 *The magnetometer.*—The variation of magnetization of the specimen under examination was measured by the deflections of a simple conventional astatic magnetometer, as described by Blackett (1952), which was mounted directly above the furnace.

By rotating the specimen through 180° about a vertical axis it was possible to measure the variation of the magnitude of the horizontal component of magnetization of the specimen with temperature. The sensitivity of the magnetometer was such that a specimen with a horizontal magnetic moment equal to $3{\cdot}6 \times 10^{-4}$ c.g.s. if placed at a distance of 4 cm from the lower magnet, and at right angles to it, produces a scale deflection of 1 cm.

4.2 *The furnace.*—The furnace consists of water jacket, insulating material, and a non-inductively wound heating element of nichrome wire. All materials apart from the heating element are non-magnetic.

The specimen is introduced into the furnace on the end of a silica tube, and is held in a ceramic specimen holder. The thermocouple used for measuring the temperature passes through the centre of the silica tubing, and because it must be of non-magnetic material, one of Pt –Pt 13 per cent Rh is used. The thermocouple junction is inserted into a hole bored in the base of the specimen. The specimen can be raised into position, and also rotated beneath the furnace by means of a cord and pulley system. Heating was carried out in an inert atmosphere of N_2, introduced through the base of the furnace.

4.3 *The Helmholtz coil system.*—In order to provide a field free space, first to reduce the magnetic control on the magnet system, and secondly to provide a zero field in which heating takes place, the whole apparatus is surrounded by a Helmholtz coil system. The coils each contain 96 turns of 22 S.W.G. enamelled single cotton covered wire and are of radii 56 cm and 52 cm respectively. The currents required to annul the vertical component of 0·56 Oe and the horizontal component of 0·22 Oe were 390 and 135 mA respectively. Along the axis of these coils the value of the compensating field does not vary by more than 1/500 of its value at the centre of the system for 10 cm on either side of the centre point.

5. Standard thermal demagnetization curves

To ensure that the internal temperature of a specimen does not lag behind the external temperature of the specimen, and that the heating is uniform, a series of tests was carried out using a basalt specimen to the side of which a thermocouple was attached. Another thermocouple was inserted into a hole bored into the specimen. For the final heating procedure used, the temperature of the inside of the specimen fell behind the outside temperature by only 5 °C.

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Thermal demagnetization curves were obtained using artificial specimens of magnetite, nichol and hematite. Each of these substances was powdered and mixed into plaster of Paris to simulate the typical specimen used, that is a cylinder of diameter 2·2 cm and height 2·2 cm. The Curie point of magnetite is 578 °C, of nichol is 360 °C and of hematite is 680 °C, and these values were obtained using the thermal demagnetization apparatus, thus providing a check of the temperature measurement.

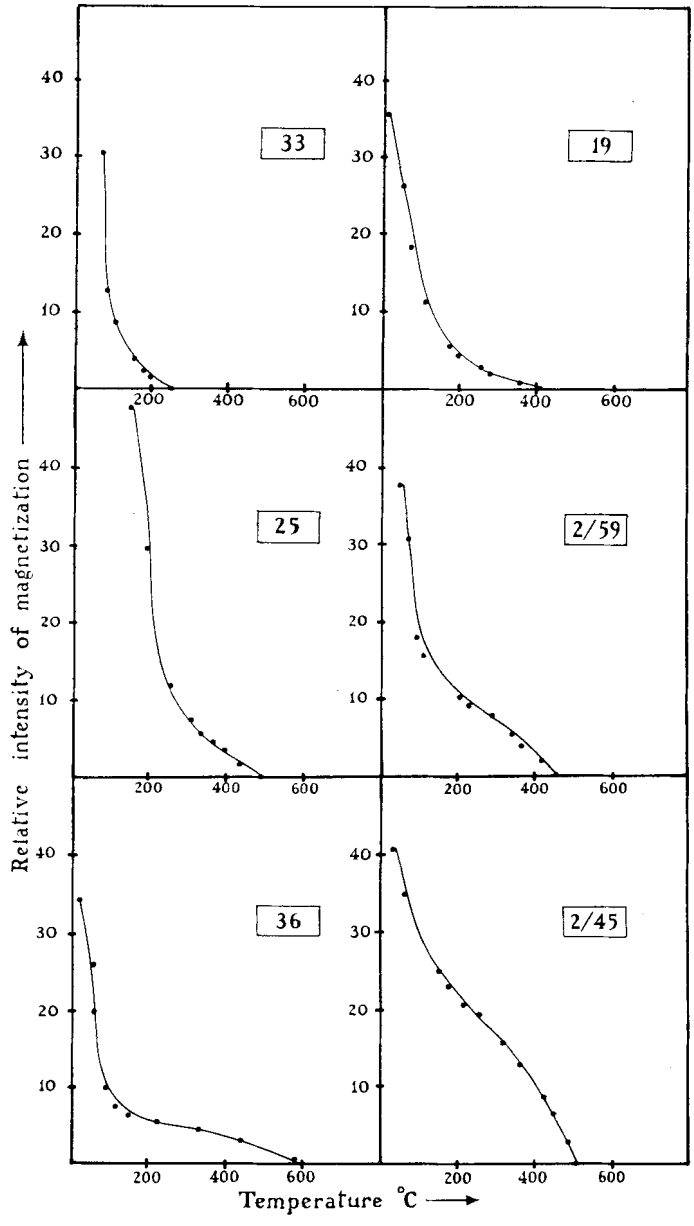


FIG. 3.—Typical thermal demagnetization curves for Older Volcanic basalts.

6. Results of the thermal demagnetization experiments

Typical thermal demagnetization curves are shown in Figure 3. It is observed from these curves that there are both single-phase and multi-phase magnetic

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systems present in these basalts, and the shapes of the thermal demagnetization curves obtained, and the temperatures at which final demagnetization occurs varies widely from specimen to specimen.

These results suggest that the magnetic mineral in these basalts is a titano-magnetite, the composition of which varies considerably from one locality to the next. An X-ray diffraction photograph of one of the specimens revealed that the magnetic mineral was cubic with a unit cell size of 8·44 Å compared with 8·39 Å for magnetite. This expanded unit cell dimension also suggests strongly that the specimen contains titano-magnetite.

In Figure 3 are given the thermal demagnetization curves for specimens 33, 36, 19, 25, 2/45 and 2/59. Specimens 19, 2/45 were reversely magnetized rocks so that the normal and reversed rocks are similar to the extent that they both contain single- and multi-component magnetic systems.

It is also interesting to observe that specimens 19, 36 were among those which exhibited large changes in direction of magnetization during the 9 months of storage.

The thermal demagnetization curve for a cylinder from specimen 25 the disks from which did not change their directions of magnetization very much during the 9 months' storage is also given in Figure 3. The specimen with the more stable thermal demagnetization properties, that is with a composition more nearly that of magnetite seems to possess the more stable magnetization during storage.

7. A.C. demagnetization experiments

Because of the observed instability in these basalts, the author constructed a.c. demagnetization apparatus similar to that described by Irving & others (1961). It was thought that with this apparatus it would be possible to remove unstable

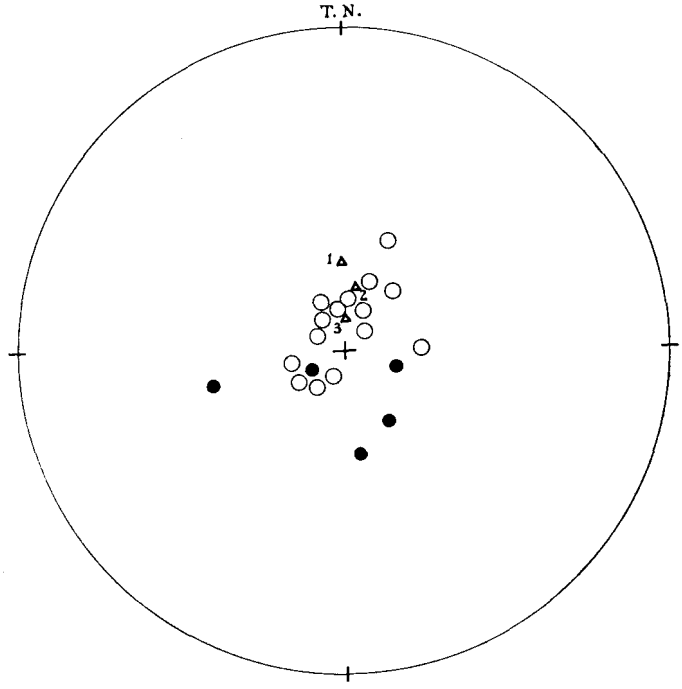


FIG. 4.—Stereographic projection showing mean site directions after a.c. cleaning. Conventions used are as in Figure 1.

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unwanted secondary components of magnetization which were observed to be present to some degree in these rocks. Such temporary components of magnetization would affect the mean direction of magnetization of the site directions obtained from the measurement of 2 to some unknown extent. Using the method suggested by Irving & others (1961) specimens of Older Volcanic basalt from each of the sites sampled were cleaned in progressively higher a.c. magnetic fields to determine the distribution with least scatter for the site in question.

In Figure 4 are given the mean site directions after this cleaning was carried out. Table 1 lists the mean direction of magnetization. Only 20 sites are listed after cleaning in Table 4 as site number 6 was found to be very unstable.

The mean direction of magnetization has increased in inclination, which is to be expected if temporary components in the direction of the Earth's field have been acquired by the rocks.

Figure 5 shows the position of the ancient pole relative to Australia, as determined from the results given in Table 1. This agrees very closely with the pole position for Tertiary basalts which have been discussed by Irving & others (1961) (See Table 2).

Table 2

Older Volcanics	Pole Position Latitude	Longitude
	deg.	deg.
Green & Irving (1958)	66·8 S	122·7 E
Present investigation—after washing	63 S	140 E
Tertiary Volcanics		
Irving & others (1961)	63 S	137 E

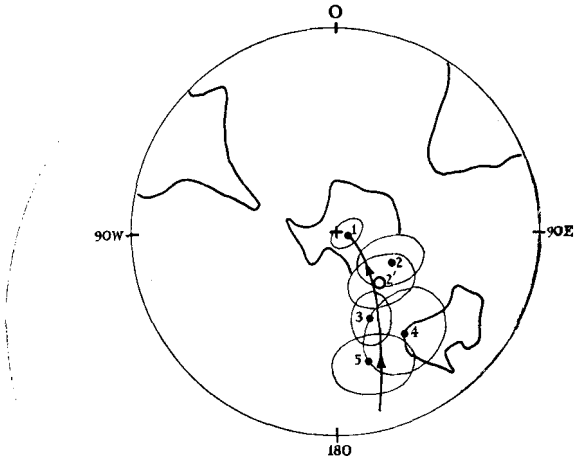


FIG. 5.—Ancient pole relative to Australia. The pole positions are numbered as follows.

1. Newer Volcanics of Victoria.
2. Older Volcanics of Victoria as determined by Irving and Green.
2'. Older Volcanics of Victoria after cleaning to remove unstable components of magnetization.
3. Tasmanian dolerite sites.
4. Kuttung Volcanics.
5. Kuttung Sediments.

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7.2.—The results of further a.c. demagnetization experiments carried out by the author are shown in Figures 6 and 7. These diagrams show the changes in the direction of N.R.M. of two basalt specimens, 2/45 and 2/59, after cleaning

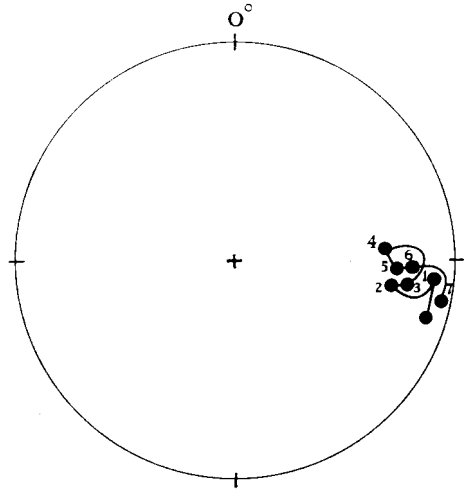


FIG. 6.—Stereographic projection of the change in direction of magnetization of specimen 2/59 on a.c. demagnetization.

1. 45 Oe
2. 90
3. 150
4. 300
5. 450
6. 600
7. 750

O north-seeking direction plotted on upper hemisphere.
● north-seeking directions plotted on lower hemisphere.

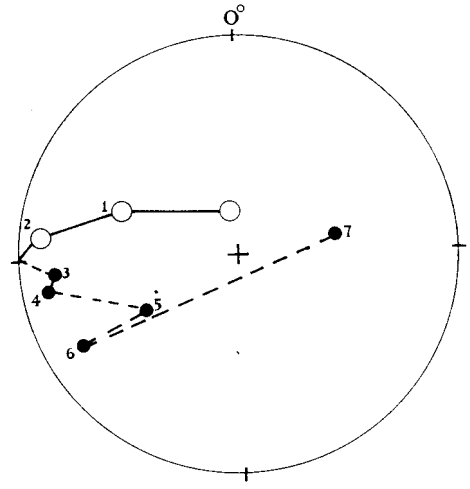


FIG. 7.—Stereographic projection of change in direction of magnetization of specimen 2/45 on a.c. demagnetization.

1. 45 Oe
2. 90
3. 150
4. 300
5. 450
6. 600
7. 750

Convention as in Figure 6.

in progressively higher a.c. fields. The thermal demagnetization curves for these two specimens are shown in Figure 3. Grain size analysis of these two rocks

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reveal that they both contain iron minerals which have similar spectra of grain sizes, and these are shown in Figures 8 and 9. The only difference to be observed then is that of composition, that is, specimen 2/45 consists mainly of a high Curie point titano-magnetite, 2/59 mainly of a low Curie temperature titano-magnetite.

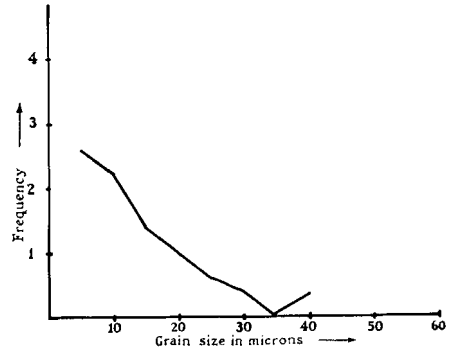


FIG. 8.—Figure showing the results of grain size analysis of iron ore minerals in specimen 2/59.

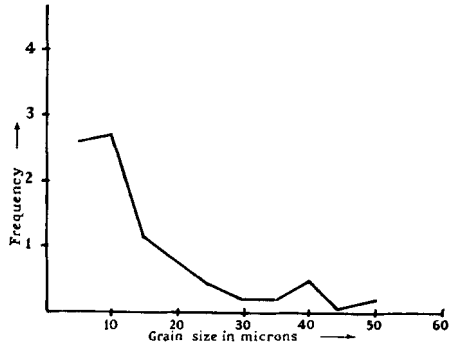


FIG. 9.—Figure showing the results of grain size analysis of iron ore minerals in specimen 2/45.

7.3.—It has been observed by Irving & others (1961) that in many cases the directions of magnetization observed in specimens taken from a single lava flow have a scatter which is usually greater than the experimental errors involved, and that among other causes, the presence of small temporary components of magnetization could explain this.

In Table 3 are given the values of precision K (Fisher (1953)) for the distributions of the directions of magnetization of specimens taken from Abberfeldie Quarry near Melbourne, before and after they have been washed in a.c. fields of 86 and 150 Oe.

Table 3

Peak cleaning field		K
Original distribution		34·6
86	Oe	450
173	Oe	82

The scatter of the directions of magnetization is greatly reduced through the a.c. demagnetization and these results indicate that in this case the large scatter

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in direction is due to temporary components of magnetization of varying magnitude being acquired by the rocks.

7.4.—In Figure 3 is shown the thermal demagnetization curves for specimen 33. In Figure 10 is the thermal demagnetization curve for the same specimen after it has been held at 400–500 °C for a period of three hours in an atmosphere of air. Oxidation similar to that described by Akimoto & Katsura (1958) has no doubt occurred and resulted in the formation of a titanomagnetite of higher Curie temperature. Similar changes were reported by Parry (1960).

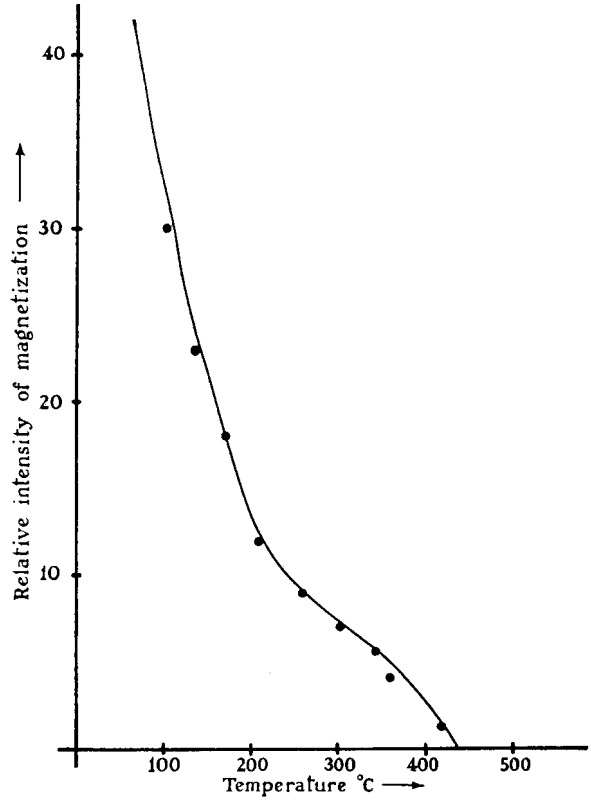


FIG. 10.—Thermal demagnetization curve for specimen 33 after it has been held at 400–500 °C for 3 hours in air and allowed to acquire a T.R.M. in 0·6 Oe.

To show that this was the case and it was not the formation of perhaps an impure hematite that took place, some of the magnetic fraction from specimen 33 which was subjected to the heat treatment was separated and examined by X-ray diffraction methods. A powder photograph showed that the magnetic material was still a titanomagnetite in that it has a slightly expanded cell size compared to magnetite ($\simeq$ 8·42 Å). No hematite lines could be detected.

Figure 11 shows the a.c. demagnetization curves for specimen 33 bc before and after it has been held at 400–500 °C in air for the period of one hour. In Figures

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12, 13 are the changes in direction of magnetization for the same specimen 33 bc at the various points at which cleaning was carried out. It is seen that the magnetization that the specimen possesses after heating is more stable, and retains its direction at higher cleaning fields. This apparently results from the formation of a titanomagnetite of higher Curie temperature leading to increased stability of magnetization. Such dependence on composition has also been described in Section 7.2.

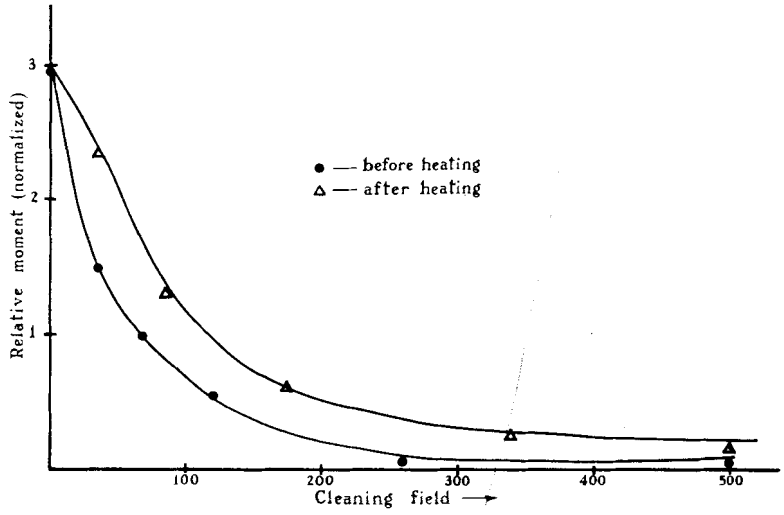


FIG. 11.—Figure showing the change in a.c. demagnetization properties for specimen 33 bc after it has been heated in air for 1 hour.

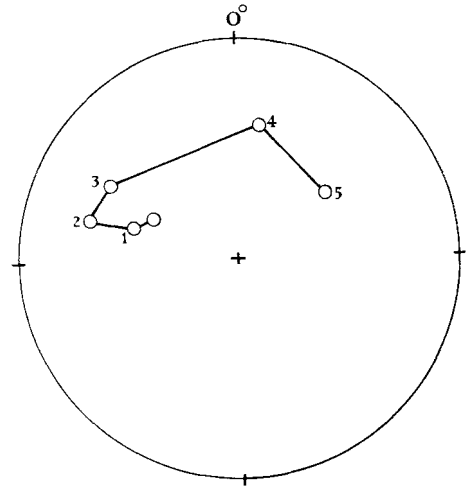


FIG. 12.—Stereographic projection showing the change in direction of magnetization of specimen 33 bc on a.c. demagnetization.

1. 35 Oe
2. 86
3. 173
4. 346
5. 519

Convention as used in Figure 6.

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8. Conclusions

The investigation of the palaeomagnetism of the Old Volcanic basalts of Victoria has led to some interesting results concerning their stability.

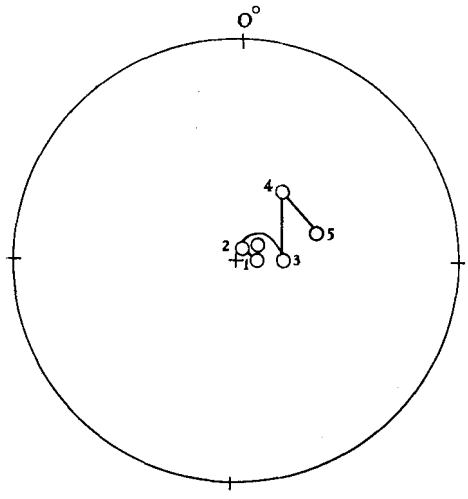


FIG. 13.—Stereographic projection showing the change in direction of magnetization of specimen 33 bc on cleaning, after it has been held at 400–500 °C in air for 1 hour.

1. 35 Oe
2. 86
3. 173
4. 346
5. 519

Convention as used in Figure 6.

The presence of, for the most part, small temporary components of magnetization in these rocks has been detected, and by removal of these unstable components it has been shown that the basalts are magnetized with an inclination of magnetization slightly greater than was previously reported. The change in direction is 4°.

This result follows from a more detailed study than that of Green and Irving, and supersedes their result. Also it is noteworthy that the pole position is almost the same as that of other Tertiary Volcanics reported by Irving & others (1961).

Actually much greater instability has been observed in these basalts, and this has been reported previously by Mumme (1962). In this investigation, one of the sites of mixed polarity of magnetization, which was reported by Irving & Green (1958) to contain both normal and reversed rocks, was studied in detail. It was shown (by a.c. cleaning methods) that the normally magnetized rocks in this site (Table 4, No. 2) were caused by temporary components of magnetization in the direction of the present Earth's field, superimposed on a stable reversed direction, for the normal rocks became reversely magnetized after a.c. treatment.

There also seems to be a dependence of stability of magnetization on the composition of the titano-magnetite contained in any particular basalt, suggested by the observations of 6, 7.2, and 7.4.

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Table 4

This table contains primary data.

S is the number of samples collected from each site. ω_1 is the number of disks cut from these samples.
R_1 is the resultant. D_S is the declination east of magnetic north.
I_S is the inclination, positive downwards. Reversed, normal and mixed polarities are signified by R, N, and X respectively.

Site	Locality	Specimen No.	S	Before cleaning D_S(M, N)	I_S	ω_1	R_1	Polarity	After cleaning D_S(M, N)	I_S	ω_1	R_1	Polarity
1	Bayview Quarry	1, 2	2	22	−62	13	12·91	N	26	−61	3	2·99	N
2	Quarry at Drouin South	3, 4, 5	3	−38	−77	19	18·48	X	266	+71	3	2·64	R
3	Jindivick Quarry	6, 7, 8	3	8	−76	31	25·97	N	9	−72	6	5·63	N
4	Tarago Quarry	9, 10	2	−17	−61	18	17·86	N	−6	−67	4	3·99	N
5	Jacob's Creek Quarry	11, 12, 13	3	18	−62	10	9·45	N	6	−62	6	5·83	N
6	S.E.C. Quarry North of Yallourn	14, 15, 16	3	17	−73	25	24·1	N					
7	Paradise Creek Quarry	17, 18	2	16	−44	15	14·8	N	10	−47	4	3·94	N
8	Hillside Quarry	19, 20, 21	3	64	+26	26	24·85	R	67	+47	6	5·95	R
9	True Blue Quarry	22, 23	2	342	+62	8	6·58	R	50	+79	4	3·27	R
10	Quarry on Berries Creek Road	24, 25, 26	3	299	+51	22	21·80	R	311	+63	6	5·97	R
11	Quarry at Leongantha	27, 28, 29	3	−19	−72	19	18·98	N	−15	−73	6	5·93	N
12	Evans Quarry	30, 31, 32	3	72	−55	15	11·53	N	20	−77	6	5·69	N
13	West of Nobbies near Jetty	33, 34, 35	3	−24	−63	6	5·97	N	216	−79	2	1·99	N
14	West of Nobbies	36, 37	2	−74	−67	19	17·46	N	243	−75	4	3·78	N
15	West of Nobbies	38, 39, 40	3	289	−72	23	19·2	N	220	−82	6	5·58	N
16	East of Nobbies	41, 42, 43	3	40	−59	25	21·62	N	71	−63	6	5·66	N
17	Kitty Millers Beach	44, 45	2	87	−73	13	10·92	N	246	−75	4	3·74	N
18	Pyramid Rock	46	1	330	+53	6	5·93	R	338	+56	2	1·99	R
19	Jennings Quarry	47, 48, 51	3	14	−73	25	23·26	N	−30	−72	6	5·87	N
20	Beach near Jennings Quarry	49, 50	2	188	−74	14	13·31	N	−32	−74	4	3·99	N
21	Cape Schanck	52, 53	2	−21	−68	10	9·75	N	−41	−77	4	3·89	N

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References

Akimoto, S. & Katsura, T., 1958. *J. Geomag. & Geoelect.*, **10,** 69.
Blackett, P. M. S., 1952. *Phil. Trans. Roy. Soc. A*, **245,** 303.
Fisher, R. A., 1953. *Proc. Roy. Soc., A*, **217,** 295.
Green, R. & Irving, E., 1958. *Proc. Roy. Soc. Victoria*, **70,** Part 1.
Irving, E. & Green, R., 1957. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7,** 6.
Irving, E. & others, 1961. *Phil. Mag.*, **6,** 62, 225.
Mumme, W. G., 1962. *Geophys. J.*, **6,** 546.
Parry, L. G., 1960. *J. Geophys. Res.*, **65,** 242.
Watson, G. S., & Irving, E., 1957. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7,** 6.

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